

Ryan Marin

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EDUCATION

University of Cambridge, *Applied Mathematics and Theoretical Physics, Ph.D.*

- Strings & ML theory under Dr. Sven Krippendorff

2026 – 2030
Cambridge, UK

Princeton University, *Physics, A.B.*

- GPA: 3.7/4.0

- Awards: Manfred Pyka Memorial Physics Prize (x2):

- Relevant Coursework: Statistical Mechanics, Stochastic Processes, Differential Geometry, Complex Analysis, Algebraic Geometry, Quantum Field Theory, String Theory, Black Holes & High Energy Gauge Theories, General Relativity, Algebraic Topology

2021 – 2025
Princeton, NJ

San Juan Hills High School, *General Studies, Valedictorian*

- GPA/ACT/SAT: 4.000/4.924, 36/36, 1580/1600

2017 – 2021
Orange County, CA

FINANCE

Jane Street, *Quantitative Trading Intern*

- Over the course of 11 weeks, developed two projects on Options and Domestic ETF desks
- Attended daily courses on algorithms, machine learning, and live trading

2024
New York City, NY

Tiger Capital Management, *Analyst; Technology (2021), Industrials & Energy (2022)*

- Produced company models, DCFs, wrote and pitched stock evaluations

- Balanced portfolio for oldest, most competitive (<3%) fund at Princeton (>150k AUM)

2021 – 2023
Princeton, NJ

RESEARCH

Nonfactorization in AdS_2 Quantum Gravity, *Senior Thesis*

- Wrote a 122-page review of modern advances in low-dimensional quantum gravity, JT gravity and its adaptation as a coset-model $SL(2, \mathbb{R})$ BF theory, interpreting the Saad-Shenker-Stanford ensemble model and its connections to modern complexity theory.
- Mentored by Juan Maldacena, *Carl P. Feinberg Professor*, Institute for Advanced Study

2024 – 2025
Princeton, NJ

C^k -regular extremal black holes in maximally-symmetric spacetime and the third law of black hole thermodynamics, *Undergraduate Research, Junior Paper*

- Adapted Kehle–Unger "null-cone gluing" to curved spacetime, proving the first construction of extremal Reissner–Nordström black holes from smooth Cauchy data, providing a counter-example and analysis to the classical "third law" of black-hole thermodynamics in dS_4/AdS_4
- Written under guidance of Mihalis Dafermos, preprint published on Arxiv/2411.17938

2023 – 2024
Princeton, NJ

Toward a resolution of the black hole information paradox: the quasi-analytic transition between self-gravitating strings and black holes, *Undergraduate Research, Junior Paper*

- Literature review of self-gravitating string solutions within M-theory landscape
- Interpreted Witten-Maldacena-Chen entropy methods and Horowitz-Polchinski mechanism
- Written under Nissan Itzhaki, School of Natural Sciences, Institute for Advanced Study

2024
Princeton, NJ

Myers-Perry 4+1 black holes via superradiance and null geodesic trapping, *Undergraduate Research*

- Researched stability of Myers-Perry black holes within landscape of high-dimensional black hole solutions via PDE modeling of null geodesic trapping and superradiance
- Mentored by Dr. Gabriele Benomio, Princeton University

2023
Mission Viejo, CA

MISCELLANEOUS

Technical Skills: Python, Java, LaTeX, Mathematica, MATLAB, Excel, Machining & Soldering

Languages: English [Native], French [B2], Mandarin Chinese [B1], Akkadian [A2]

Interests: Linguistics, Rock Climbing, Music Theory, Metaphysics